

O-C plug-in

Overview

This plug-in computes **O-C** (observed minus computed) values for times of light-curve extrema and displays an O-C diagram. A fixed test ephemeris (period and epoch) is compared against observed times of maximum (or minimum) light to reveal timing errors.

Primary reference for the six-clock O-C tutorial (Tables 13.1–13.2): AAVSO *Variable Star Astronomy*, chapter 13 — freely available PDF: [Variable Stars and O–C Diagrams](#) (Grant Foster). The same material appears in Foster, *Analyzing Light Curves*, ch. 13.

Two plug-ins are provided.

Plug-in	Menu	Role
O-C	Tools → O-C...	Main O-C tool (includes Export CSV... in results)
O-C demo data	File → O-C demo data...	Optional Foster clock light curves

The O-C tool can be opened from the **Tools** menu, even when no observations have been loaded, since other than a **From observations** option, an **Imported timings file** option allows a previously exported/created timings file to be opened instead.

Published help (PDF): [O-C on aavso.github.io](#).

Installation

1. In VStar, choose **Tools** → **Plug-in Manager...**
2. Select and install **O-C** (required). Optionally install **O-C demo data** for Foster tutorial light curves.
3. **Close and restart VStar** so new menu items appear.

See also [AAVSO VStar Plug-in Library](#).

Optional: **File** → **Preferences...** → **Plug-in Settings** to check the plug-in download URL matches your VStar version (see the [VStar wiki installation recipes](#)).

Quick start

1. **Tools** → **O-C...**
2. Set **data source**, ephemeris (period and epoch), and other options in the parameter dialog.
3. Click **OK** (a file chooser opens if you chose **Imported timings file**).
4. Review the results dialog: **O-C diagram**, **Data table**, and **Fit summary**.

See **Examples** for Foster demo walkthroughs. **Appendix A** covers all six clocks (demo light curves or imported timings). For a typical workflow on your own AID data (measure → export CSV → re-import), see **Appendix C**.

Examples

These three walkthroughs use the optional **O-C demo data** plug-in (no real star or downloaded timing file required). Install it if needed (see **Installation**), then restart VStar. Each example uses the VSA/Foster test ephemeris **P = 1.0 d**, **epoch = 2450000.0** (Tables 13.1–13.2 in the [Chapter 13 PDF](#)).

Example	Demo scenario (VSA clock)	Purpose
1	Foster clock 4 — epoch jump	Two-segment fit with parallel segments
2	Foster clock 5 — period change	Compare linear, quadratic, and two-segment (different slopes)
3	Foster clock 6 — slowing clock	Compare linear vs quadratic for a curved O-C

Clocks 1–3 (correct ephemeris, epoch offset, period error) are covered in **Appendix A**. Demo menu labels keep the “Foster clock” wording used in the plug-in; the numbers and tables are those of the VSA chapter.

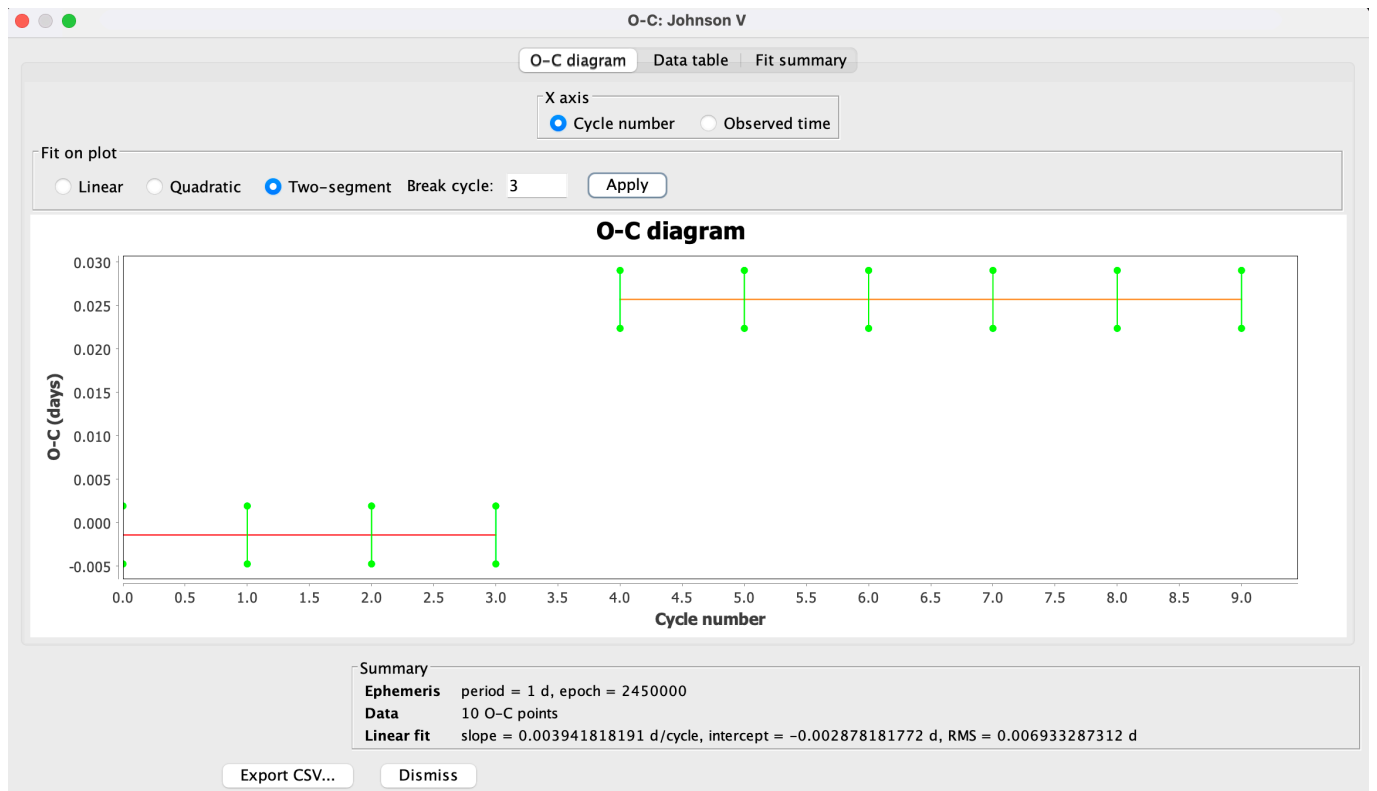
Shared setup for each example (details only in Example 1; abbreviated later):

1. **File** → **O-C demo data...** → choose the demo scenario for that clock → **OK**.
2. **Tools** → **O-C...** → **From observations** → **Star metadata** → **Event**: Maximum → **Timing method**: Parabolic interpolation → **OK** → select **Johnson V**.

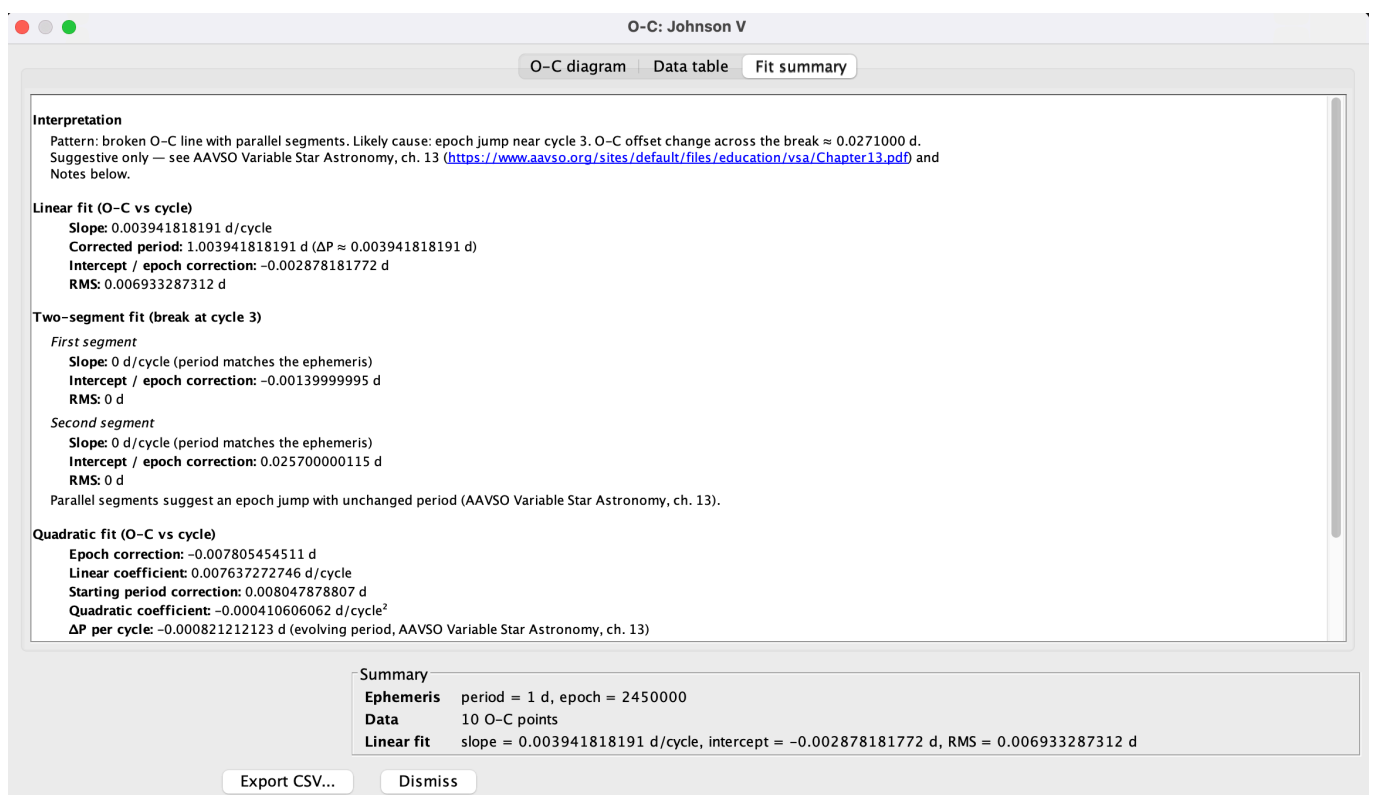
Example 1 — Epoch jump (clock 4)

A sudden step in O-C with unchanged slope on each side (VSA Table 13.2, clock 4).

1. Load **Foster clock 4 — epoch jump** and open O-C as in the shared setup. Note the load-message hint: break cycle **3** (cycles 0–3 are the pre-jump plateau; cycle 4 is already post-jump).
2. On the **O-C diagram**, the points step upward at cycle 4. Select **Two-segment**, enter **Break cycle 3**, click **Apply**. Both segments should be nearly flat (slope ≈ 0) with different intercepts.



- Open **Fit summary**. The **Interpretation** should describe parallel segments (epoch jump with unchanged period).

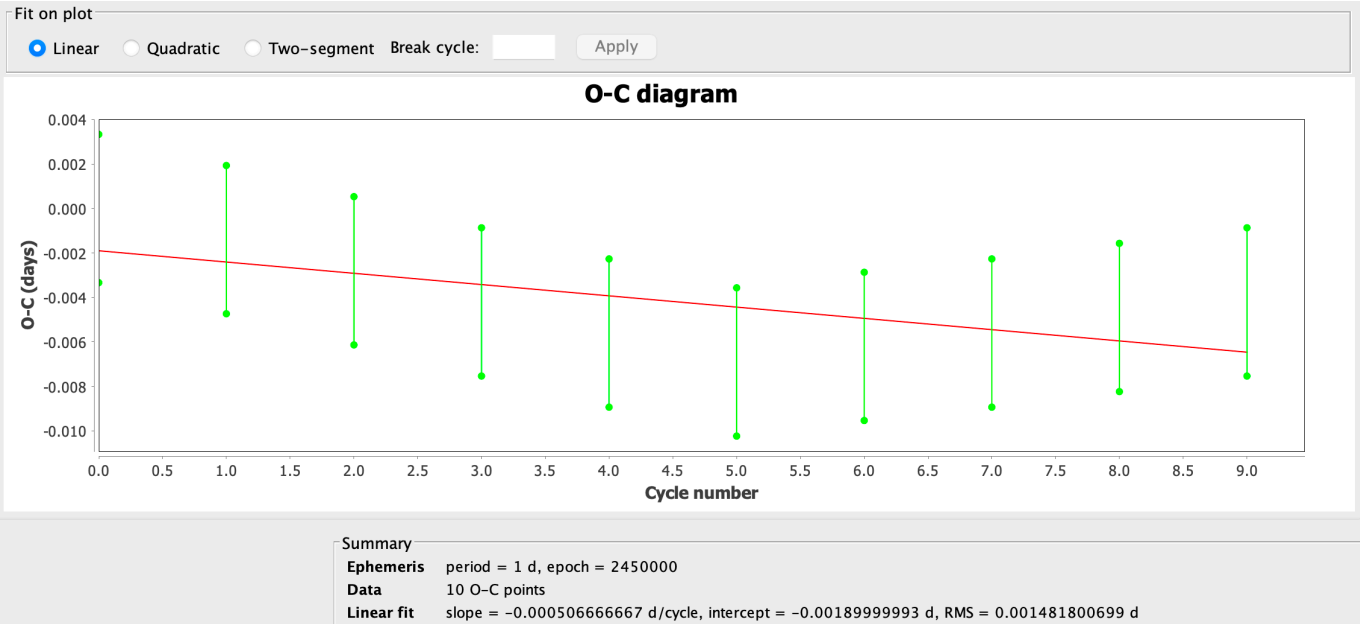


Example 2 — Period change (clock 5)

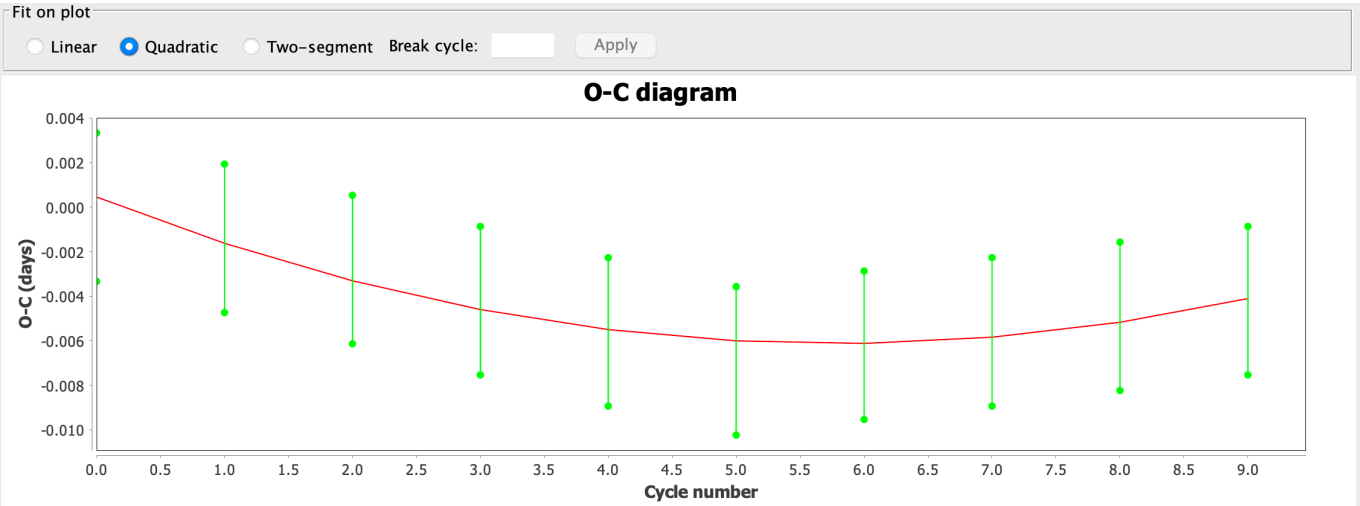
A slope change mid-run: a single line or parabola is a poor model; two segments with different slopes match

Table 13.2 (clock 5).

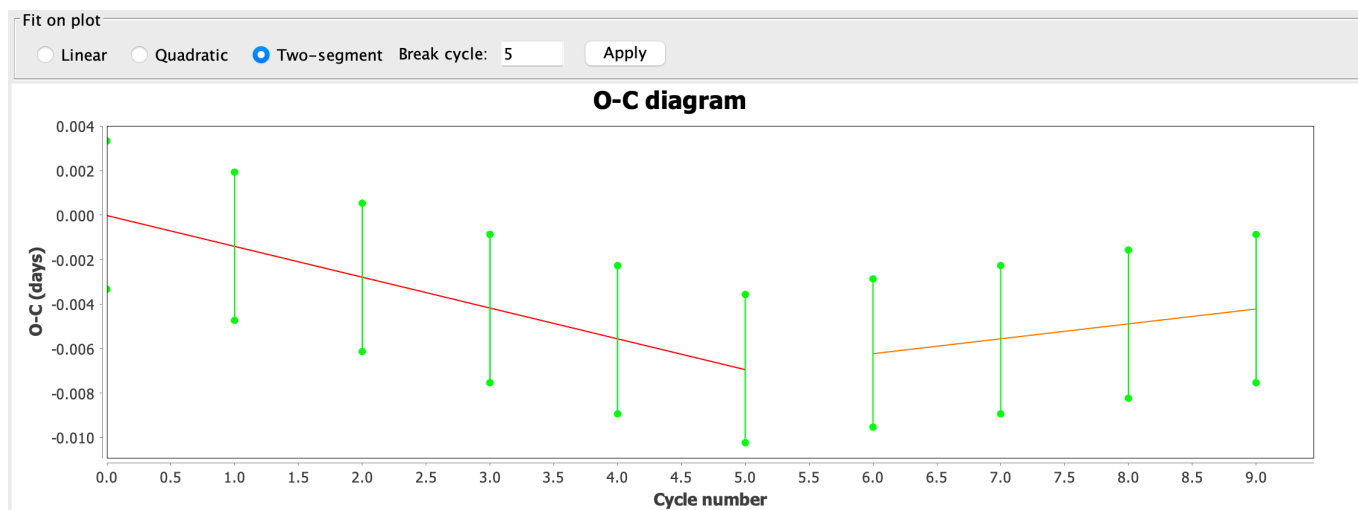
1. Load **Foster clock 5 — period change** and open O-C as in the shared setup. Note the hint: break cycle 5 (cycles 0–5 are the first slope; cycle 6 starts the new regime).
2. Leave **Fit on plot** on **Linear**. The points show a kink; one line is only a compromise across both segments.



3. Switch to **Quadratic**. A parabola may look somewhat better, but it still does not capture a sharp period change at one cycle.



4. Switch to **Two-segment**, enter **Break cycle 5**, click **Apply**. Separate lines before and after the break should show different slopes.



- Open **Fit summary**. With **Two-segment** selected, the **Interpretation** should report a period change. Switch **Fit on plot** back to **Linear** or **Quadratic** and return to **Fit summary** — the text follows the fit shown on the diagram. Compare with Example 1 (parallel segments / epoch jump).

Interpretation

Pattern: broken O-C line with different slopes. Likely cause: period change near cycle 5. Segment period change ≈ 0.0020557 d/cycle (second minus first segment). Suggestive only — see AAVSO Variable Star Astronomy, ch. 13 (<https://www.aavso.org/sites/default/files/education/vsa/Chapter13.pdf>) and Notes below.

Linear fit (O-C vs cycle)

Slope: -0.000506666667 d/cycle
Corrected period: 0.999493333333 d ($\Delta P \approx -0.000506666667$ d)
Intercept / epoch correction: -0.001899999993 d
RMS: 0.001481800699 d

Two-segment fit (break at cycle 5)

First segment

Slope: -0.00138571426 d/cycle
Corrected period: 0.99861428574 d ($\Delta P \approx -0.00138571426$ d)
Intercept / epoch correction: -0.000019047587 d
RMS: 0.000028171761 d

Second segment

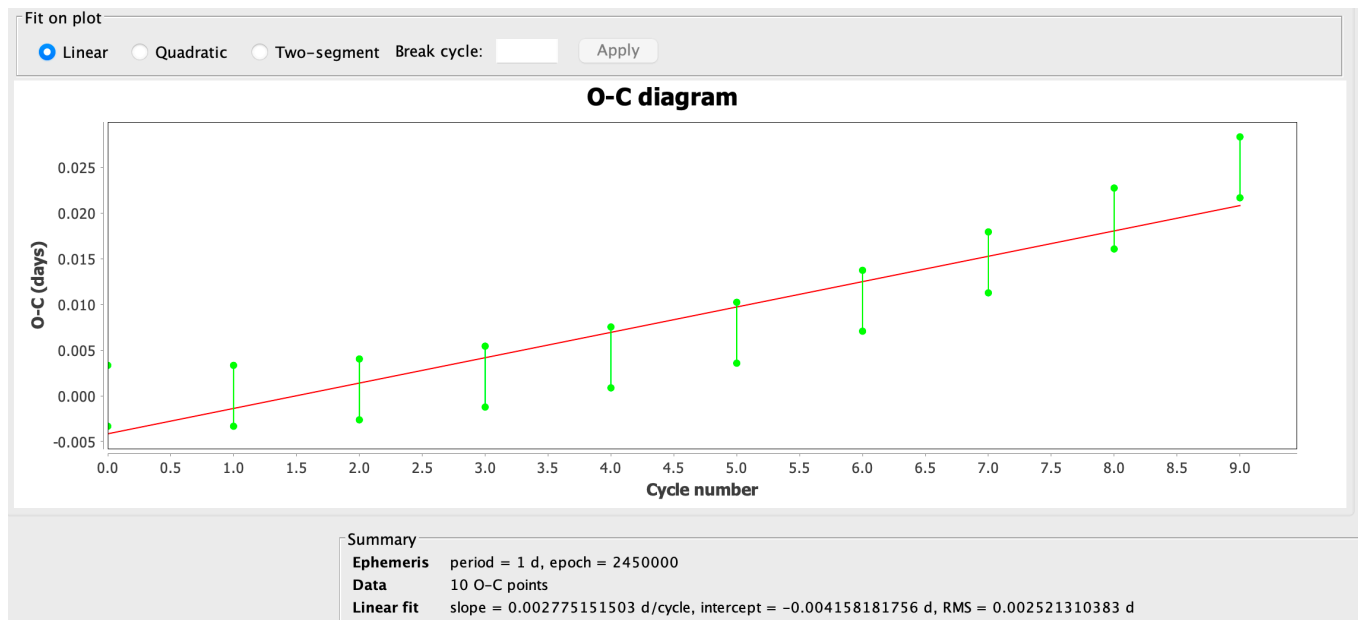
Slope: 0.000670000073 d/cycle
Corrected period: 1.000670000073 d ($\Delta P \approx 0.000670000073$ d)
Intercept / epoch correction: -0.010250000516 d
RMS: 0.000027386039 d

Different slopes suggest a period change near cycle 5 (AAVSO Variable Star Astronomy, ch. 13).

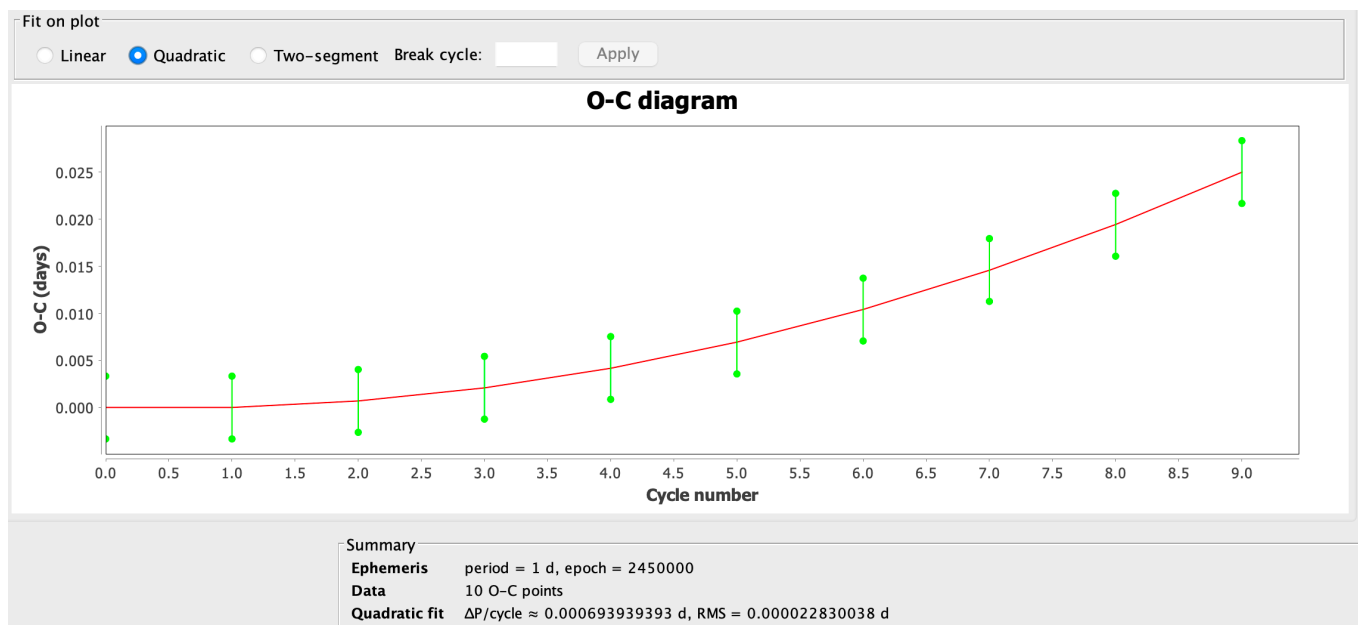
Example 3 — Evolving period (clock 6)

A smoothly curving O-C from an evolving period (Table 13.2, clock 6). Here **linear** vs **quadratic** is the main contrast (no break cycle required).

- Load **Foster clock 6 — slowing clock** and open O-C as in the shared setup.
- Leave **Fit on plot** on **Linear**. A straight line cannot follow the curve.



3. Switch to **Quadratic**. The parabola should track the points.



4. **Fit summary** should interpret a curved O-C (evolving period) when **Quadratic** is selected.

Interpretation

Pattern: curved (parabolic) O–C. Likely cause: evolving period. $\Delta P/\text{cycle} \approx 0.0006939$ d. Suggestive only — see AAVSO Variable Star Astronomy, ch. 13 (<https://www.aavso.org/sites/default/files/education/vsa/Chapter13.pdf>) and Notes below.

Linear fit (O–C vs cycle)

Slope: 0.002775151503 d/cycle
Corrected period: 1.002775151503 d ($\Delta P \approx 0.002775151503$ d)
Intercept / epoch correction: –0.004158181756 d
RMS: 0.002521310383 d

Quadratic fit (O–C vs cycle)

Epoch correction: 0.000005454604 d
Linear coefficient: –0.000347575768 d/cycle
Starting period correction: –0.000694545464 d
Quadratic coefficient: 0.000346969697 d/cycle²
 ΔP per cycle: 0.000693939393 d (evolving period, AAVSO Variable Star Astronomy, ch. 13)
RMS: 0.000022830038 d

Notes

Caution (AAVSO Variable Star Astronomy, ch. 13): for long-period variables and other stars with cycle-to-cycle period scatter, O–C residuals are not white noise — apparent trends may reflect intrinsic period jitter rather than a true ephemeris change. See also Foster (1993, JAAVSO 22, 145) on O–C autocorrelation. Chapter PDF: <https://www.aavso.org/sites/default/files/education/vsa/Chapter13.pdf>

O–C pattern reference: [AAVSO Variable Star Astronomy, ch. 13](#)

Further reading — X Tri, Z Tau, and other real stars

The VSA chapter discusses real variables (e.g. **X Tri**, **Z Tau**, SU Vir) with O–C diagrams built from published **times of extremum**, not re-measured light curves. Bundled fixtures:

Star	Role	File
X Tri (EB)	VSA Table 13.9 minima	xtri_table_13_9.txt
Z Tau (Mira)	AAVSO LPV times of maximum	zttau_maxima.txt

Downloads: [xtritable13_9.txt](#), [zttau_maxima.txt](#).

Reproduce Activity 13.6 (X Tri):

1. Optionally load **X Tri** from the AAVSO International Database with a JD range covering Table 13.9 minima ($\approx 2442720 - 2450090$), Visual or any band you prefer. The light curve is context only (e.g. for vertical markers); the O–C points come from the imported timings file, not from re-timing this sparse photometry.
2. **Tools** → **O–C...** → **Imported timings file** → choose `xtri_table_13_9.txt`.
3. Period **0.975352** d and epoch **2442502.721** fill from the file's `# period=..., epoch=...` comment when present; set **Event** to **Minimum**.
4. Inspect O–C vs cycle (and try **Quadratic** on Fit on plot). Expect a clear long-term trend — the teaching linear ephemeris is **not** a perfect flat O–C; that is the point of the exercise.

Z Tau (long-period / Mira maximum O–C):

1. Optionally load **Z Tau** from AID with **JD** $\approx 2417000 - 2458100$ (roughly 1905–2018, spanning the bundled maxima). Visual is fine for context; sparse Mira data are not meant for auto-timing this

exercise.

2. **Tools** → **O-C...** → **Imported timings** → `ztau_maxima.txt` (source: [AAVSO Maxima & Minima of LPVs](#)).
3. Period **466.2** d and epoch from the file header fill automatically; set **Event** to **Maximum**.
4. Expect a **curved / multi-segment** O–C: Z Tau’s period has evolved (historical means ~500 d early, GCVS later ~466 d, VSX often ~446 d). A fixed trial **P** is deliberately imperfect—the diagram is the lesson.

Do **not** expect to match these diagrams by auto-timing sparse AID Visual photometry: they use **published times of extremum**. For other stars, use literature timings or measure dense photometry, then **From observations / Edit timings**.

Clock demos (Tables 13.1–13.2) remain in **Examples / Appendix A**. For an AID export/re-import outline on real photometry, see **Appendix C**.

For **eclipsing binaries**, decades of **eclipse timings** can reveal **apsidal motion** (sinusoidal O-C from general relativity and tidal distortion) or **period change** from mass transfer; those effects need long baselines.

Results dialog

After O-C is computed, a modeless results window opens with three tabs:

Tab	Contents
O-C diagram	O-C points (optional error bars); X axis and Fit on plot controls
Data table	Cycle, observed time, computed time, O-C, uncertainty, obs count, QC
Fit summary	Pattern interpretation for the selected fit, plus fit details

Fit on plot — radio buttons choose which fit is drawn:

Option	When available	Notes
Linear	≥2 O-C points	Default; single least-squares line
Quadratic	≥3 O-C points	Curved trend (Foster clock 6 / evolving period)
Two-segment	≥4 O-C points (≥2 per segment)	Enable Break cycle + Apply on the same row

Fit summary — leads with an **Interpretation** paragraph for the fit currently selected on the diagram (VSA / Foster ch. 13 patterns). Fit coefficients and the LPV period-scatter caution follow below.

Light-curve check — solid vertical **O** markers (observed timing) and lighter dashed **C** markers (ephemeris

prediction) are drawn on the raw-data light curve. Re-running O-C replaces previous O-C markers; loading a new star clears them. Labels (cycle numbers) appear only when there are ≤ 30 markers.

Edit timings — at the top of the results dialog:

Control	Action
Place O on light curve	Click the raw LC to place a free JD as an observed time. Chart pan/zoom drag is disabled while this is on.
Snap to nearest observation	When placing or dragging, snap O to a nearby observation (when a series is available).
Remove selected	Delete the selected table row / O marker (Delete key).
Clear all	Remove every timing.
Drag O marker	Press near an O vertical and drag to a new free JD.
Data table O (time)	Editable; type a new JD and the O-C list rebuilds.

C markers stay ephemeris-driven (not editable). QC notes show `manual` , `snap` , or the automatic method (e.g. KvW). Export CSV reflects the current edited list.

For **Kwee–van Woerden** seeds, the summary still reports how many cycles were timed versus skipped before editing; per-point $\sigma(\text{O-C})$ / **QC** may change when you replace automations with free JDs.

Export CSV... on the results dialog saves O-C points and fit metadata to a file.

Components (summary)

O-C tool

Parameter dialog uses a compact two-column layout. **Ephemeris source** is a drop-down only (not typable); edit **Period** and **Epoch** in their fields.

Data source options:

- **From observations** — automatic timing methods on a selected series.
- **Imported timings file** — load O times from a file.
- **Edit timings on light curve** — open an empty editor (period/epoch required) and place/remove O markers interactively. Optionally pick a series for snap.

After **OK**, choose an observation series when needed; the results dialog opens next. Use **Export CSV...** there to save results.

O-C demo data (observation source)

Optional. Loads synthetic light curves whose **maximum timings** match Tables 13.1–13.2 of the [VSA Chapter 13 PDF](#) (clocks 1–6). Bump shape is identical for every clock; only timing differs. Then run **Tools** → **O-C...** with **From observations** and the suggested ephemeris from the load message ($P = 1\text{ d}$).

Appendix A — Validation against VSA Tables 13.1–13.2

The six test clocks (theory $P = 1\text{ day}$, epoch = cycle 0) are from AAVSO *Variable Star Astronomy* chapter 13 ([PDF](#)). Computed time $C_n = n$. $O-C = O_n - C_n$ (Table 13.2). The same tables appear in Foster, *Analyzing Light Curves*, ch. 13.

Reference O-C values (Table 13.2)

Cycle	Clock 1	Clock 2	Clock 3	Clock 4	Clock 5	Clock 6
0	0	+0.0035	0	−0.0014	0	0
1	0	+0.0035	+0.0021	−0.0014	−0.0014	0
2	0	+0.0035	+0.0042	−0.0014	−0.0028	+0.0007
3	0	+0.0035	+0.0062	−0.0014	−0.0042	+0.0021
4	0	+0.0035	+0.0083	+0.0257	−0.0056	+0.0042
5	0	+0.0035	+0.0104	+0.0257	−0.0069	+0.0069
6	0	+0.0035	+0.0125	+0.0257	−0.0062	+0.0104
7	0	+0.0035	+0.0146	+0.0257	−0.0056	+0.0146
8	0	+0.0035	+0.0167	+0.0257	−0.0049	+0.0194
9	0	+0.0035	+0.0188	+0.0257	−0.0042	+0.0250

Demo scenario mapping

VSA / Foster clock	Demo menu label	Expected in plug-in
1	Foster clock 1 — correct ephemeris	Flat O-C ≈ 0
2	Foster clock 2 — epoch offset	Flat O-C $\approx +0.0035$ d
3	Foster clock 3 — period error	Slope $\approx +0.0021$ d/cycle
4	Foster clock 4 — epoch jump	Step at cycle 4; break 3
5	Foster clock 5 — period change	Slope change; break 5
6	Foster clock 6 — slowing clock	Curved O-C; select Quadratic on Fit on plot

Method A — imported timings

For each clock, download the timing file (links below), then:

1. **Tools** → **O-C...** → **Imported timings file**.
2. **Period:** 1.0 d, **Epoch:** 2450000.0.
3. Compare O-C and **Fit summary** to Table 13.2.

Method B — demo observation source

See **Examples** for clocks 4–6. For any clock:

1. **File** → **O-C demo data...** → pick a Foster clock (labels match the demo scenario mapping table above).
2. **Tools** → **O-C...** → **From observations**; use the suggested ephemeris from the load message (**Star metadata** pre-fills period and epoch).
3. Compare O-C, **Fit on plot**, and **Fit summary** to Table 13.2.

Timing files (Table 13.1 → times)

Epoch **2450000.0**, period **1.0** d for O-C tool entry. Download from aavso.github.io (published with plug-in docs; same files in `plugin/doc/foster/` in the repo).

Clock	Scenario	Download
1	Correct ephemeris	foster_clock1.txt
2	Epoch offset	foster_clock2.txt
3	Period error	foster_clock3.txt
4	Epoch jump (try break cycle 3)	foster_clock4.txt
5	Period change (try break cycle 5)	foster_clock5.txt
6	Slowing clock (Quadratic on Fit on plot)	foster_clock6.txt

Timing files — published extrema (X Tri, Z Tau)

File	Contents	Ephemeris
xtritable13_9.txt	122 published minima (cycle + JD); repeated cycles = different observers	(P = 0.975352 d), (t_0 = 2442502.721)
ztau_maxima.txt	84 AAVSO LPV times of maximum (cycle + JD)	(P = 466.2 d), (t_0 = 2415246.2)

Star	Optional AID JD load (context LC)	Event
X Tri	$\approx$ 2442720 – 2450090	Minimum
Z Tau	$\approx$ 2417000 – 2458100	Maximum

See **Further reading** above.

Appendix B — Reference

Data source in the parameter dialog is either **From observations** or **Imported timings file**. Period and epoch define the test ephemeris (required; same time system as the data — see below).

Ephemeris source

When **From observations** is selected:

UI label	Meaning
Phase plot	Period and epoch from the active phase plot (View → Phase Plot, or Period Analysis → New Phase Plot)
Star metadata	Period and epoch from the loaded star record (often from an AID download or other source that supplies catalogue fields)
Manual entry	Type period and epoch yourself

Disabled for **Imported timings** — enter period and epoch directly, unless the file is an O-C export CSV (metadata is read from the file).

Imported timings file format

One timing per line: `cycle time [sigma]` or `time [sigma]`. Whitespace, comma, or semicolon separators. `#` comments allowed.

Use **one consistent time system** for observations, epoch, and imported timings (JD, HJD, BJD_TDB, etc.). The plug-in does not convert between systems; mix them and O-C will be wrong. Homogenise with VStar's HJD/BJD tools first if needed.

You can also re-open a file saved with **Export CSV...** from a previous O-C run. The tool reads timings from the data rows and, for export files, fills in **Period**, **Epoch**, and **Event** from the `# period=..., epoch=...` comment and the `Event` column (leave those fields blank in the parameter dialog). Export columns are `O_time` and `C_time` (not a specific JD flavour).

Timing methods (From observations)

Observations only (with **Extreme N%**, **KvW folds**, and **Minimum observations per cycle** as applicable):

- **Parabolic interpolation** — parabolic refinement at discrete extremum (default; good for smooth peaks such as the Foster demos).
- **Mean JD of extreme N%** — mean time of brightest (or faintest) N% in each cycle.
- **From current model function** — analytic extrema from a JD-based model.
- **Kwee–van Woerden** — single-eclipse time-of-minimum/maximum (Deeg 2020), via the same library as **Tools** → **Kwee–van Woerden....** After cycle bucketing, O-C trims an eclipse-like window around the discrete extremum (half-width capped at $\sim 0.2 P$), requires at least **7** points in that window, then runs KvW. Failed or undersampled cycles are **skipped** (no silent fall-back to parabolic). Use **Minimum** for eclipsing binaries, a good trial period / epoch so one eclipse sits per cycle bin, and inspect the vertical markers. Recommended min observations per cycle: ≥ 7 .

Event type

- **Maximum** or **Minimum**
- **Both maxima and minima** — for eclipsing binaries on real data (not used in six-clock tutorial in the VSA Chapter 13 PDF)

Interpreting O-C diagrams

Pattern	Likely cause
Flat at 0	Ephemeris matches
Flat, offset	Epoch wrong, period OK
Linear slope	Period wrong
Broken line, parallel segments	Epoch jump
Broken line, different slopes	Period change
Curved (parabolic)	Evolving period → select Quadratic on the diagram



Appendix C — AID workflow (export and re-import)

Typical path on your own data: measure extrema from a loaded light curve, export timings, then re-open them without reprocessing the AID.

Suggested target: **RZ Cas** (AUID `000-BBF-490`), Algol-type EA. Period ≈ 1.20 d (VSX). **Suggested JD range:** **2452700 – 2461000** (about 22 years); load **V** only.

1. Load **RZ Cas** from the AAVSO International Database with the JD range and **V** series above.
2. **Tools** → **O-C...** → **From observations** → **Star metadata** → **Event:** Minimum → **OK** → select **V**.
3. Review the diagram, then **Export CSV...** (e.g. `rz_cas_timings.csv`).
4. **Tools** → **O-C...** again → **Imported timings file** → select the CSV. Period, epoch, and event are taken from the export metadata when present.

O-C

Data source: From observations ⌵

Ephemeris source: Star metadata ⌵

Period (days): 1.19525556

Epoch (HJD): 2460646.2029

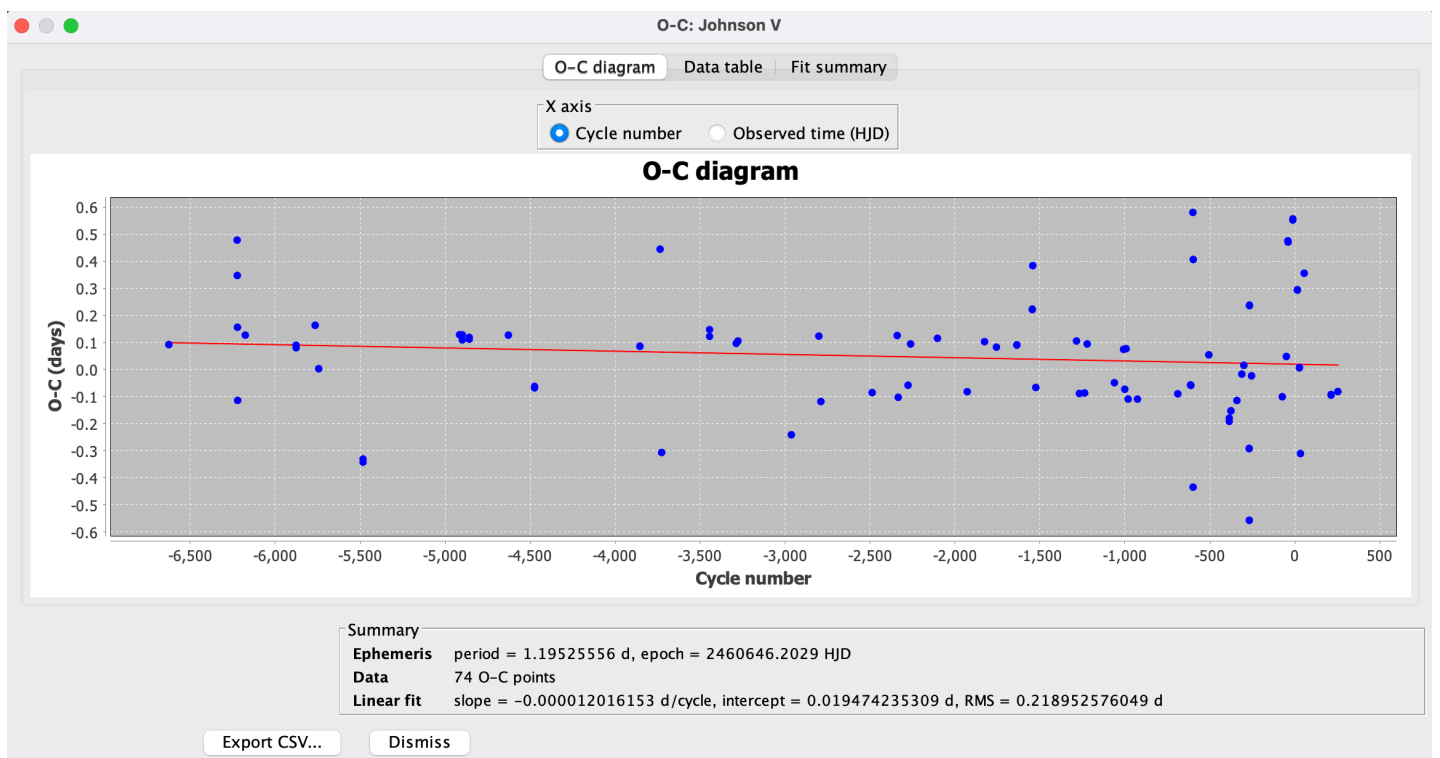
Event: Maximum light (minimum magnitude) ⌵

Timing method: Parabolic interpolation ⌵

Extreme N% (mean timing method): 10

Minimum observations per cycle: 3

Help
Cancel
OK



References

- AAVSO *Variable Star Astronomy*, chapter 13 (Grant Foster) — free PDF: [Variable Stars and O-C Diagrams](#) (primary reference for Tables 13.1–13.2 and the six-clock examples)
- Foster, G. *Analyzing Light Curves*, O-C Analysis, ch. 13, pp. 263–282 (same clock material; book form)
- [VStar plug-in library](#)
- [VStar issue #93](#)
- Deeg, H. J. (2020). The Kwee–van Woerden method revisited. *Research Notes of the AAS* / arXiv:2011.09231 — used by the KvW plug-in and O-C KvW path

- [Kwee–van Woerden plug-in](#) (standalone tool; S(T) / mirror QC remains there)

Version history

Pre-release development is tracked in [issue #93](#) and git commits on branch

`93-implement-o-c-analysis-plugin` . A numbered release changelog will start with the **first published plug-in version**.